

Submission D solution to Problem 10

Problem Statement

Prove the following statement:

Let Γ be a finite simplicial graph with $|\mathcal{V}(\Gamma)| \geq 3$, and (M_v, φ_v) be a separable tracial von Neumann algebra for each $v \in \mathcal{V}$. If Γ is irreducible (in the sense that there does not exist subgraphs Γ_1, Γ_2 such that $\mathcal{V}(\Gamma_1) \cup \mathcal{V}(\Gamma_2) = \mathcal{V}(\Gamma)$ and every $(v, u) \in \mathcal{E}(\Gamma)$ for all $v \in \mathcal{V}(\Gamma_1), u \in \mathcal{V}(\Gamma_2)$), and each M_v has a trace zero unitary, then the graph product von Neumann algebra M is properly proximal.

Solution

Theorem 1. *Let Γ be a finite simplicial graph with $|\mathcal{V}(\Gamma)| \geq 3$, and (M_v, φ_v) be a separable tracial von Neumann algebra for each $v \in \mathcal{V}(\Gamma)$. If Γ is irreducible (its complement graph Γ^c is connected), and each M_v has a trace zero unitary, then the graph product von Neumann algebra $M_\Gamma = \overline{\bigotimes_\Gamma M_v}$ is properly proximal.*

Proof. We proceed by mathematical induction on the number of vertices $n = |\mathcal{V}(\Gamma)|$.

Step 1: Base Cases ($n = 3$)

Let Γ be an irreducible graph on 3 vertices. Irreducibility dictates that the complement graph Γ^c is connected. The only connected graphs on 3 vertices are the complete graph K_3 and the path P_3 . We rely on the foundational dimension criteria for proper proximality of free products (Boutonnet, Ioana, and Peterson, 2021, Theorem 4.1): The free product of two tracial von Neumann algebras $A * B$ is properly proximal provided $\dim A \geq 2$ and $\dim B \geq 3$. By hypothesis, each vertex algebra M_v possesses a trace-zero unitary, guaranteeing $\dim M_v \geq 2$.

- **Case 1a** ($\Gamma^c = K_3$): The graph Γ is totally disconnected. Thus, $M_\Gamma = M_1 * M_2 * M_3$. We rewrite this via associativity as the free product of $A = M_1$ and $B = M_2 * M_3$. Since $\dim M_2 \geq 2$ and $\dim M_3 \geq 2$, their free product contains a diffuse subalgebra (isomorphic to the group von Neumann algebra of $\mathbb{Z}_2 * \mathbb{Z}_2$), yielding an infinite linear dimension, $\dim B = \infty \geq 3$. Since $\dim A \geq 2$, the free product $A * B$ strictly satisfies the threshold and is properly proximal.
- **Case 1b** ($\Gamma^c = P_3$): Let the complement edges be $(1, 2)$ and $(2, 3)$. The graph Γ strictly contains only the single edge $(1, 3)$. Thus, the graph product evaluates as $M_\Gamma = (M_1 \overline{\otimes} M_3) * M_2$. We set $A = M_1 \overline{\otimes} M_3$ and $B = M_2$. The tensor product yields $\dim A = \dim M_1 \times \dim M_3 \geq 2 \times 2 = 4 \geq 3$. Since $\dim B \geq 2$, the free product $A * B$ strictly satisfies the threshold and is properly proximal.

Step 2: Inductive Amalgamated Decomposition

Assume the theorem holds for all irreducible graphs with k vertices, where $3 \leq k < n$. Let Γ be an irreducible graph with $n \geq 4$ vertices. Since Γ^c is a connected graph with at least 4 vertices, it must contain a non-cut vertex v . Thus, the induced subgraph $\Gamma' = \Gamma \setminus \{v\}$ remains irreducible and has $n - 1 \geq 3$ vertices. By the induction hypothesis, the subalgebra $A = M_{\Gamma'}$ is properly proximal.

Let $L = \text{lk}_\Gamma(v)$ denote the link of v in Γ . In the complement graph $G = \Gamma^c$, the vertices of L are exactly the non-neighbors of v . Because v must have at least one neighbor in the connected graph G , the set L is strictly smaller than $\mathcal{V}(\Gamma')$. The full graph product M_Γ natively decomposes into the amalgamated free product:

$$M_\Gamma = A *_C B, \quad \text{where } C = M_L \text{ and } B = C \overline{\otimes} M_v.$$

Step 3: Geometric Ping-Pong on $\mathbb{B}(L^2 M_\Gamma)$

By Boutonnet, Ioana, and Peterson (2021), a tracial von Neumann algebra M is properly proximal if there does not exist an M -central state φ on $\mathbb{B}(L^2 M)$ extending the trace τ_M such that $\varphi(T) = 0$ for all operators $T \in \mathcal{J}_{\text{prop}}(M)$, where $\mathcal{J}_{\text{prop}}(M)$ is the norm-closed proper proximal ideal generated by $\{xe_D^M y \mid x, y \in M, D \subset M \text{ an amenable von Neumann subalgebra}\}$.

Assume for the sake of contradiction that M_Γ is not properly proximal. Then there exists such an M_Γ -central state φ on $\mathbb{B}(L^2 M_\Gamma)$ that completely vanishes on $\mathcal{J}_{\text{prop}}(M_\Gamma)$.

Let $L^2 M_\Gamma$ decompose into the orthogonal direct sum of $L^2 C$ and the spaces of reduced alternating words over the orthocomplements $A \ominus C$ and $B \ominus C$. Define Q_A as the orthogonal projection onto the closed span of all alternating words whose leftmost letter belongs to $A \ominus C$. Define Q_B as the orthogonal projection onto the span of all alternating words whose leftmost letter belongs to $B \ominus C$. The alternating word basis strictly enforces the exact orthogonal equality $1 = e_C + Q_A + Q_B$.

Let $u_v \in M_v$ be the given trace-zero unitary. Define $u = 1 \otimes u_v \in C \overline{\otimes} M_v = B$. Since $\tau(u_v) = 0$, we have $E_C(u) = 0$, ensuring $u \in B \ominus C$. Left multiplication of any alternating word starting with $A \ominus C$ by u strictly increases the word length by 1, producing a valid alternating word that begins with $u \in B \ominus C$. Thus, $u \operatorname{ran}(Q_A) \subset \operatorname{ran}(Q_B)$. Since u is unitary, this ensures the projection inequality $u Q_A u^* \leq Q_B$. Applying the M_Γ -central state φ , we deduce $\varphi(Q_A) = \varphi(u Q_A u^*) \leq \varphi(Q_B)$.

Because v is a non-cut vertex of $G = \Gamma^c$, we select specific vertices in G to construct orthogonal shifts. Let $N_G(v)$ be the neighbors of v in G . Elements in $N_G(v)$ are connected to v in G , meaning they are strictly disconnected from v in Γ , so they do not belong to L (hence $M_{N_G(v)} \not\subset C$).

- **Case A** ($|N_G(v)| \geq 2$): Choose distinct $x, y \in N_G(v)$.

If $x \sim y$ in G ($x \asymp y$ in Γ): Set $w_1 = u_x$ and $w_2 = u_x u_y u_x$. Because $x \notin L$, $E_C(w_1) = 0$, so $w_1 \in A \ominus C$. In the graph product, any reduced word containing at least one letter outside L is strictly orthogonal to M_L . Since $x \asymp y$ in Γ , u_x and u_y do not commute, meaning $u_x u_y u_x$ is a reduced word. Because it contains $x \notin L$, its conditional expectation onto C is exactly $E_C(w_2) = 0$, so $w_2 \in A \ominus C$. Further, $w_2^* w_1 = u_x^* u_y^* u_x^* u_x = u_x^* u_y^*$. This is a reduced word with letters outside L , guaranteeing $E_C(w_2^* w_1) = 0$.

If $x \asymp y$ in G ($x \sim y$ in Γ): Set $w_1 = u_x$ and $w_2 = u_y$. Both are strictly in $A \ominus C$. Since they commute, $w_2^* w_1 = u_y^* u_x$. Because $x \neq y$ and both $\notin L$, this is a product of mutually commuting elements strictly outside L , guaranteeing $E_C(w_2^* w_1) = E_C(u_y^*) E_C(u_x) = 0$.

- **Case B** ($|N_G(v)| = 1$): Let x be the unique neighbor. Since $|V(G)| \geq 4$, x must have another neighbor $y \neq v$ in G . Because x is the unique neighbor of v in G , $y \notin N_G(v)$, meaning $y \in L$ (thus $M_y \subset C$). Set $w_1 = u_x$ and $w_2 = u_x u_y u_x$. Then $w_1 \in A \ominus C$. For w_2 , since $x \sim y$ in G ($x \asymp y$ in Γ), u_x and u_y do not commute. The word $u_x u_y u_x$ is

strictly reduced in the graph product and contains $x \notin L$, enforcing $E_C(w_2) = 0$, so $w_2 \in A \ominus C$. Furthermore, $w_2^*w_1 = u_x^*u_y^*$. This is a reduced word containing $x \notin L$, guaranteeing $E_C(w_2^*w_1) = 0$.

In all cases, we rigorously secure unitaries $w_1, w_2 \in A \ominus C$ satisfying $E_C(w_2^*w_1) = 0$. For any $\xi, \zeta \in \text{ran}(Q_B)$, they are spanned by alternating words whose first letter is in $B \ominus C$. Left-multiplying by $w_1 \in A \ominus C$ strictly increases the alternating word length by 1, mapping ξ to an alternating word starting with $A \ominus C$. Thus, $w_1Q_Bw_1^* \leq Q_A$. Moreover, orthogonality is strictly preserved: $\langle w_1\xi, w_2\zeta \rangle = \tau(\zeta^*w_2^*w_1\xi) = 0$. This holds because $E_C(w_2^*w_1) = 0$, ensuring that the middle product $w_2^*w_1 \in A \ominus C$, so $\zeta^*(w_2^*w_1)\xi$ remains a valid alternating word of length ≥ 3 , which necessarily has trace zero.

Thus, $w_1Q_Bw_1^*$ and $w_2Q_Bw_2^*$ are mutually orthogonal subprojections of Q_A . Applying φ , we acquire $2\varphi(Q_B) = \varphi(w_1Q_Bw_1^*) + \varphi(w_2Q_Bw_2^*) \leq \varphi(Q_A)$. Combined with the established $\varphi(Q_A) \leq \varphi(Q_B)$, this algebraically forces $\varphi(Q_A) = 0$ and $\varphi(Q_B) = 0$. Consequently, $\varphi(e_C) = \varphi(1 - Q_A - Q_B) = 1$. Because $C \subset A$, the projection geometry strictly demands $e_C \leq e_A$, enforcing $\varphi(e_A) = 1$.

Step 4: The Algebraic Lifting Contradiction

Define the functional ψ on $\mathbb{B}(L^2A)$ by $\psi(T) = \varphi(e_ATe_A)$, utilizing the exact spatial identification $\mathbb{B}(L^2A) \cong e_A\mathbb{B}(L^2M_\Gamma)e_A$. Because $\varphi(e_A) = 1$, ψ is a valid state on $\mathbb{B}(L^2A)$. For any $a \in A$, a commutes with e_A , ensuring $\psi(aT) = \varphi(e_AaTe_A) = \varphi(ae_ATe_A) = \varphi(e_ATe_Aa) = \psi(Ta)$. Thus, ψ is strictly A -central. Moreover, since $\varphi(e_A) = 1$, we unconditionally have $\varphi(1 - e_A) = 0$. The Cauchy-Schwarz inequality for states ensures $|\varphi(a(1 - e_A))|^2 \leq \varphi(aa^*)\varphi(1 - e_A) = 0$, rigorously guaranteeing $\psi(a) = \varphi(e_Aae_A) = \varphi(ae_A) = \varphi(a) = \tau_M(a) = \tau_A(a)$, meaning ψ extends the standard trace on A .

We now evaluate ψ on the proper proximal ideal $\mathcal{J}_{\text{prop}}(A)$. By definition, $\mathcal{J}_{\text{prop}}(A)$ is the norm-closed ideal in $\mathbb{B}(L^2A)$ generated by operators of the form Xe_D^AY , where $X, Y \in A$ and $D \subset A$ is an amenable von Neumann subalgebra. Under the spatial identification, e_D^A corresponds precisely to the restriction of e_D^M , so on L^2M_Γ it acts strictly as $e_D^Me_A = e_Ae_D^Me_A$. For any generator Xe_D^AY with $X, Y \in A$, we evaluate:

$$\psi(Xe_D^AY) = \varphi(e_AXe_D^MYe_A).$$

Since $X, Y \in A$ and A is a von Neumann subalgebra, the orthogonal projection e_A strictly commutes with X and Y (which inherently act by left mul-

multiplication on $L^2 M_\Gamma$), allowing us to rewrite the argument as $Xe_A e_D^M e_A Y = Xe_D^M Y$. Thus, $\psi(Xe_D^A Y) = \varphi(Xe_D^M Y)$.

Because $D \subset A \subset M_\Gamma$ is an amenable subalgebra of M_Γ and $X, Y \in A \subset M_\Gamma$, the operator $Xe_D^M Y$ inherently belongs to the proper proximal ideal $\mathcal{J}_{\text{prop}}(M_\Gamma)$ strictly by definition. By our founding contradiction assumption, φ vanishes entirely on $\mathcal{J}_{\text{prop}}(M_\Gamma)$, which rigorously dictates $\varphi(Xe_D^M Y) = 0$. By linearity and norm-continuity, ψ completely vanishes on all of $\mathcal{J}_{\text{prop}}(A)$.

Thus, ψ is an A -central state on $\mathbb{B}(L^2 A)$ extending τ_A that definitively vanishes on $\mathcal{J}_{\text{prop}}(A)$. However, by the induction hypothesis, A is properly proximal, mathematically prohibiting the existence of exactly such a state. This is a direct and fatal contradiction. Therefore, no such state φ can exist, definitively establishing that the graph product von Neumann algebra M_Γ is properly proximal. $\square$